

2023 AMC 10A Problem 2 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10A Problem 2. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10A Problem 2

**Problem:**

The weight of $\frac{1}{3}$ of a large pizza together with $3\frac{1}{2}$ cups of orange slices is the same as the weight of $\frac{3}{4}$ of a large pizza together with $\frac{1}{2}$ cup of orange slices. A cup of orange slices weighs $\frac{1}{4}$ of a pound. What is the weight, in pounds, of a large pizza?

Answer Choices:

A. $1\frac{4}{5}$

B. 2

C. $2\frac{2}{5}$

D. 3

E. $3\frac{3}{5}$

Solution:

Let p be the weight of a large pizza and q be the weight of a cup of orange slices, in pounds. Then

$$\frac{1}{3}p + \frac{7}{2}q = \frac{3}{4}p + \frac{1}{2}q$$

so $3q = \left(\frac{3}{4} - \frac{1}{3}\right)p$ and $p = \frac{36}{5}q$. Because a cup of orange slices weighs $\frac{1}{4}$ of a pound, the large pizza weighs

$$\frac{36}{5} \cdot \frac{1}{4} = \frac{9}{5} = (\text{A}) \boxed{1\frac{4}{5}} \text{ pounds.}$$

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